

Network Troubleshooting Lab 001

No Internet Connectivity

Scenario:

A user reports they are unable to access internet resources. This troubleshooting lab focuses on identifying network configuration, connectivity, DNS, and routing issues using standard network diagnostic tools.

Document Information

Category: Networking / IT Support

Lab Type: Troubleshooting Case

Environment: MacBook Pro - macOS

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Date: 10/05/2026

1. Troubleshooting Methodology

Followed a structured troubleshooting approach:

1. Identify the problem
2. Establish a theory of probable cause
3. Test the theory
4. Analyze results
5. Document findings

2. Step 001 — Network Configuration Check

Command Used

networksetup -getinfo Wi-Fi

Purpose

Used to verify the device's network configuration, including:

- IP Address
- Subnet Mask
- Router/Gateway Information

Evidence

```
Last login: Mon Oct 5 19:29:35 on ttys000
ridleyherisse@Ridleys-MacBook-Pro ~ % networksetup -getinfo Wi-Fi
DHCP Configuration
IP address: 10.0.0.287
Subnet mask: 255.255.255.0
Router: 10.0.0.1
Client ID:
IPv6: Automatic
IPv6 IP address: none
IPv6 Router: none
Wi-Fi ID: 3c:0a:38:33:b6:6d
ridleyherisse@Ridleys-MacBook-Pro ~ %
```

Findings

- The workstation successfully obtained valid network configuration details, including an IPv4 address, subnet mask, and default gateway. This confirmed that the device was properly connected to the network and able to communicate with local network resources. No issues were found within the devices basic network settings.

3. Step 002 — Connectivity Test

Command Used

ping 8.8.8.8

Purpose

Used to test communication with an external IP address and verify network connectivity.

Evidence

```
Last login: Mon Oct 5 19:29:35 on ttys000
[ridleyherisse@Ridleys-MacBook-Pro ~ % networksetup -getinfo Wi-Fi
DHCP Configuration
IP address: 10.0.0.287
Subnet mask: 255.255.255.0
Router: 10.0.0.1
Client ID:
IPv6: Automatic
IPv6 IP address: none
IPv6 Router: none
Wi-Fi ID: 3c:06:38:3b:6d:6d
[ridleyherisse@Ridleys-MacBook-Pro ~ % ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=114 time=29.612 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=36.694 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=39.146 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=114 time=38.321 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=114 time=36.688 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=114 time=35.468 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=114 time=37.191 ms
^C
--- 8.8.8.8 ping statistics ---
7 packets transmitted, 7 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 29.612/34.849/38.321/3.241 ms
[ridleyherisse@Ridleys-MacBook-Pro ~ % ]
```

Findings

The ping test returned successful responses from the external IP address, confirming that the device was able to communicate outside the local network. This verified that basic network connectivity was functioning properly.

4. Step 003 — DNS Verification

Command Used

nslookup google.com

Purpose

Used to verify DNS functionality and confirm domain name resolution.

Evidence

```
Last login: Mon Oct 5 19:29:35 on ttys000
ridleyherisse@Ridleys-MacBook-Pro ~ % networksetup -getinfo Wi-Fi
DHCP Configuration
IP address: 10.0.0.207
Subnet mask: 255.255.255.0
Router: 10.0.0.1
Client ID:
IPv6: Automatic
IPv6 IP address: none
IPv6 Router: none
Wi-Fi ID: 3c:06:30:33:b6:6d
ridleyherisse@Ridleys-MacBook-Pro ~ % ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=114 time=29.612 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=36.694 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=30.146 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=114 time=38.321 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=114 time=36.680 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=114 time=35.468 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=114 time=37.101 ms
^C
--- 8.8.8.8 ping statistics ---
7 packets transmitted, 7 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 29.612/34.849/38.321/3.241 ms
ridleyherisse@Ridleys-MacBook-Pro ~ % nslookup google.com
Server:      2001:558:feed::1
Address:     2001:558:feed::1#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.191.14

ridleyherisse@Ridleys-MacBook-Pro ~ %
```

Findings

The DNS lookup successfully resolved the domain name into an IP address, confirming that DNS resolution was functioning correctly. This verified that the device could translate domain names into network addresses.

5. Step 004 — Route Analysis

Command Used

```
tracert google.com
```

Purpose

Used to analyze the network path between the device and destination.

Evidence

```
last login: Mon Oct 5 19:29:35 on tty000
ridleyhiseriseRidley's-MacBook-Pro: % networksetup -getinfo Wi-Fi
DHCP Configuration
IP address: 10.0.0.207
Subnet mask: 255.255.255.0
Router: 10.0.0.1
Client ID:
IPv4: Automatic
IPv4 IP address: none
IPv4 Router: none
Wi-Fi ID: ID: 9c080c38c33be6dd
ridleyhiseriseRidley's-MacBook-Pro: % ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=114 time=29.612 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=36.094 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=30.146 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=114 time=38.321 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=114 time=36.680 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=114 time=36.449 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=114 time=37.101 ms
^C
8.8.8.8 ping statistics
7 packets transmitted, 7 packets received, 0% packet loss
round-trip min/avg/ stddev : 39.612/34.849/38.321 ms
ridleyhiseriseRidley's-MacBook-Pro: % nslookup google.com
Server:      2001:558:feed:1
Address:     2001:558:feed:1:1653

Non-authoritative answer:
Name: google.com
Address: 142.250.192.14

ridleyhiseriseRidley's-MacBook-Pro: % traceroute google.com
traceroute to google.com (142.251.40.238), 6 hops max, 52 byte packets
 1  10.0.0.1 (10.0.0.1)  0.728 ms  0.446 ms  0.838 ms
 2  172.27.133.226 (172.27.133.226)  20.721 ms
 3  172.27.133.226 (172.27.133.226)  19.720 ms  24.000 ms
 4  po-54-r2r82.plymouth.ma.boston.comcast.net (68.85.186.193)  19.911 ms
 5  po-54-r2r281.plymouth.ma.boston.comcast.net (68.85.186.189)  20.799 ms
 6  po-54-r2r82.plymouth.ma.boston.comcast.net (68.85.186.193)  22.472 ms
 7  po-208-rar81.plymouth.ma.boston.comcast.net (162.151.151.85)  18.972 ms
 8  po-208-rar81.plymouth.ma.boston.comcast.net (162.151.151.85)  18.827 ms
 9  po-208-rar81.plymouth.ma.boston.comcast.net (162.151.151.85)  19.072 ms
10  po-208-rar81.plymouth.ma.boston.comcast.net (162.151.151.85)  20.164 ms
11  po-208-rar81.plymouth.ma.boston.comcast.net (162.151.151.85)  21.880 ms  20.391 ms
12  be-384-rs41.needham.ma.boston.comcast.net (162.151.151.41)  23.763 ms
13  be-384-rs41.needham.ma.boston.comcast.net (162.151.151.41)  23.392 ms
14  be-3284t-4304a.newyork.ny.ibone.comcast.net (96.110.42.13)  29.288 ms
15  be-384-rs41.needham.ma.boston.comcast.net (162.151.151.41)  30.954 ms
16  be-3132-pa212.livingshaye.ny.ibone.comcast.net (96.110.34.34)  30.393 ms
17  be-3284t-4304a.newyork.ny.ibone.comcast.net (96.110.42.91)  27.259 ms
18  be-3412-pa212.livingshaye.ny.ibone.comcast.net (96.110.34.46)  27.235 ms
19  be-3312-pa212.livingshaye.ny.ibone.comcast.net (96.110.34.42)  28.835 ms * 36.952 ms
20  pnh8-rs41-danvers.ny.ibone.comcast.net (142.251.40.238)  27.856 ms * 29.728 ms
ridleyhiseriseRidley's-MacBook-Pro: %
```

Findings

The traceroute command displayed the network path between the device and the destination, showing the different hops involved in communication. The results provided visibility into how traffic traveled across the network and confirmed successful routing to the destination.

6. Conclusion

The troubleshooting process verified network configuration, external connectivity, DNS resolution, and network routing behavior using standard diagnostic tools.

7. Skills Demonstrated

- TCP/IP Fundamentals
 - Network Troubleshooting
 - DNS Troubleshooting
 - Connectivity Testing
 - Command Line Tools
 - Root Cause Analysis
 - Technical Documentation
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